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ORIGINAL INVESTIGATION

Improvement of Lower-Body Resistance-Exercise Performance With Blood-Flow Restriction Following Acute Caffeine Intake

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Purpose: To examine the effects of acute caffeine (CAF) intake on physical performance in 3 sets of unilateral knee extensions with blood-flow restriction. ***Methods:*** In a double-blind crossover design, 22 trained men ingested 6 mg·kg^{-1} of CAF or a placebo (PLA), 1 h prior to performing unilateral knee-extension exercise with blood-flow restriction until exhaustion (30% of 1 maximal repetition). ***Results:*** There was a significant difference in the number of repetitions between the CAF and PLA conditions in the first set (28.3 [5.3] vs 23.7 [3.2]; $P = .005$), second set (11.6 [3.1] vs 8.9 [2.9]; $P = .03$), and total repetitions performed across the 3 sets (44.5 [9.4] vs 35.0 [6.6]; $P = .001$). Blood lactate was also significantly different ($P = .03$) after exercise between the CAF (7.8 [1.1] mmol·L^{-1}) and PLA (6.0 [0.9] mmol·L^{-1}). In regard to pain perception, there was a difference between the CAF and PLA in the second (6.9 [1.5] vs 8.4 [1.4]; $P = .04$) and third sets (8.7 [0.4] vs 9.5 [0.6]; $P = .01$). No differences were found for perceived effort. ***Conclusion:*** Acute caffeine intake increases performance and blood lactate concentration and reduces perception of pain in unilateral knee-extension exercise with blood-flow restriction.

Keywords: nutrition, resistance training, strength training

Resistance training of moderate/high load (>60% of 1 maximal repetition—RM) is considered an integral part of any physical conditioning program to increase muscle strength or muscle mass for health in adults.[1] However, in recent years, resistance training with low loads (<30% 1RM), performed with blood flow restriction (BFR), has been suggested as being similarly effective as moderate/high-load resistance training in increasing strength and muscle mass.[2] Although the physiological mechanisms regarding the improvement of strength and muscle mass after a low-load resistance training program with BFR are still unclear,[3,4] this training model has application both for healthy, nontrained subjects at different ages[5,6] and athletes.[4] In this context, although resistance training with BFR is not significantly better than traditional moderate/high-load resistance training to improve strength and/or muscle mass, it can be added into the training routine as a variation of the exercises and may provide an alternative exercise mode that some individuals find more appealing than traditional resistance training.

In the context of resistance training to increase physical performance, in addition to training per se, it is common for athletes and recreational exercisers to use ergogenic aids[7] as a means to increase or accelerate changes in physical performance. Caffeine (CAF), for example, is one of the most widely used ergogenic aids, often employed to increase performance in exercises involving muscular strength,[8] especially when performing several repetitions until exhaustion.[9] The mechanism of action for CAF has been explained by the high affinity of CAF with adenosine receptors, inhibiting the action of this substance[10] and, consequently, reducing the perception of effort and pain.[11] Likewise, CAF can promote greater performance in the propagation of signals between the brain and neuromuscular junction,[12] acting peripherally on the ryanodine channels in the release of calcium, optimizing the process of excitation–contraction of the skeletal musculature.[13] In the context of the BFR-resistance exercise model, sustained blood flow reduction to the muscle during exercise may reduce intramuscular calcium influx,[14] which could theoretically limit the effect of CAF on the excitation–contraction coupling process in skeletal muscles. However, the possible ergogenic effects of CAF have not yet been investigated in low-intensity resistance exercise with BFR. Given there is a potential for both CAF ingestion and BFR to act independently and possibly synergistically, it is important to examine if this is the case. No study to date has examined this issue.

In this sense, the purpose of this study was to examine the effects of acute CAF intake on physical performance in 3 sets of unilateral knee extension with BFR to failure. In addition, a rating of perceived exertion (RPE), perceived pain (PP), and blood lactate concentration were analyzed.

Materials and Methods

Subjects

Sample size calculation was performed considering a difference between 2 means of 3 repetitions, an expected standard deviation of 2, statistical power of 80%, and level of significance lower than .05 in a pilot sample of this study. Thus, the minimum sample required was 16 subjects. Consequently, the sample in the current study included 22 trained men (Table 1) to account for potential drop out during the experimental procedures. The inclusion criteria to take part were nonsmokers, nonusers of dietary supplements, nonusers of anabolic steroids, the absence of muscular or metabolic problems, body mass index below 30 kg·m^{-2}, trained in bodybuilding for at least 12 months (but without experience with BFR exercises), and nonhabitual CAF users. Individuals were instructed not to engage in vigorous exercise or consume alcoholic beverages for 72 hours prior to each testing session until the end of the experiment. All participants were informed about the study procedures and possible effects of CAF intake and provided informed consent to participate.

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Table 1 General Characteristics of the Sample (N = 22)

Age, y	23.4 (4.1)
Height, cm	177.2 (3.9)
Weight, kg	76.7 (4.0)
Time of training, y	2.6 (1.1)
Level of BFR during exercise, mm Hg	131.4 (14.6)
Habitual caffeine consumption, $mg \cdot d^{-1}$	80.1 (10.4)
1RM, kg	87.2 (61.1)
30% of 1RM, kg	26.2 (2.5)

Abbreviations: BFR, blood flow restriction; 1RM, 1-repetition maximum.

The study was approved by institutional ethics of the State University of Londrina (application number 1.141.230/2015).

Experimental Design

The study employed a repeated-measures, within-subjects design and was conducted during 4 nonconsecutive days with intervals of between 48 and 72 hours. Anthropometric measurements and 1RM testing were performed during the first visit. This was followed by familiarization with the scales that were used for subjective perception of effort (RPE)[15] and pain (PP).[16] On the second visit, the 1RM retest was performed. The remaining 2 visits were assigned to the experimental sessions administered using a randomized double-blind crossover design. The subjects ingested either 1 capsule of CAF or a placebo (PLA) and, after 60 minute, performed unilateral knee extension exercise with BFR (3 sets to exhaustion, 1-minute recovery interval between sets, at an intensity of 30% 1RM). The BFR was maintained throughout the whole exercise bout (all sets and repetitions). Subjects were instructed and verbally encouraged to perform the maximum number of repetitions during each set. The repetitions were performed at a rate of 1.5 seconds (via digital metronome) for both concentric and eccentric contractions. The RPE and PP were applied after the end of each set. Blood lactate was collected after the end of the exercise. At the end of the experiment, the subjects were questioned as to whether they were able to distinguish between the 2 capsules to identify which was CAF, in order to determine efficacy of blinding and the individual perception of the effect of this substance.

Maximum Strength Test

Both 1RM tests (test and retest) were performed within a 48-hour interval using a unilateral knee-extension exercise (dominant leg) (TechnoGym®, Rome, Italy) and were determined according to methods accomplished by Seo et al.[17] A warm-up was performed with a set of 10 repetitions (~50% of predicted 1RM). Individuals were allowed up to 5 attempts to determine 1RM, with a recovery interval of 3 to 5 minutes. All subjects were instructed and verbally encouraged to perform 1 correct repetition. The load was considered maximal when the subjects performed only 1 complete repetition. The highest load obtained in either the test or retest 1RM trials was used in subsequent experimental trials. Test/retest reliability for the 1RM was performed and a high intraclass correlation coefficient was found ($R = .80$).

CAF or PLA Intake

For the experimental sessions, each subject ingested a capsule containing 6 mg of CAF per kilogram of body weight and a PLA capsule (maltodextrin) with 200 mL of water administered in a randomized order. The habitual average CAF intake of the participants was assessed through a questionnaire[18] translated to Portuguese. All participants were considered low habitual CAF users (80.1 [10.4] $mg \cdot d^{-1}$). The same list was used to instruct the individuals not to consume the same substances for 48 hours prior to testing.

Blood Flow Restriction

To elicit BFR, a cuff 18 cm wide and 90 cm long was positioned on the proximal third of the thigh. A vascular Doppler (MARTEC DV600, São Paulo, Brazil) positioned on the posterior tibial artery was used to identify the sound of the passage of blood flow. From identification of the sound, the cuff was inflated until the sound was interrupted and, at that moment, the restriction value was recorded. Cuff pressure during the experimental session was maintained at 80% of the total blood flow restriction value and was released only after the end of the final set.

Blood Lactate Collection

Blood lactate concentration was obtained at the moment of rest during the first visit to the laboratory and at an interval of up to 2 minutes after the end of the final set of each experimental session. Prior to the blood sample collection, asepsis was performed with 70% alcohol on the digital pulp of the middle finger of the right hand. The puncture was performed using disposable lancets, the drop of blood (5 μL) in suspension being applied to a specific area of the reactive strip and analyzed by means of a portable lactometer (Accutrend Plus, Roche, Mannheim, Germany).

Rating of Perceived Exertion and Perceived Pain

To measure perceived exertion and perceived muscle pain, the OMNI scale (0–10)[15] and a PP visual analog scale[16] were used, respectively. Familiarization with the scales was performed on the first day of the individuals' visit to the laboratory. Subsequently, during experimental trials, at the end of each set, the RPE and PP values were also collected.

Statistical Analyses

The Shapiro–Wilk test was used to verify the distribution of the data and the Levene test was used to verify the homogeneity of the variances. Considering the normal distribution of data, for the comparison of the number of repetitions, RPE, and PP, a repeated-measures 2-way analysis of variance was used (CAF/PLA × number of sets). The Student t test for dependent samples was applied for the blood lactate analysis. Two-way analysis of variance was used to compare the performance in the total number of repetitions, RPE, and PP, among those who identified correctly the CAF capsule and those who made a mistake about the intake of the CAF capsule. In all cases, the Tukey post hoc test was used to identify significant results. In addition, to determine the magnitude of the findings, Cohen d effect sizes (ESs) were calculated for the differences between PLA and CAF, following the classification: small ($0.20 < ES < 0.50$), medium ($0.50 \leq ES < 0.80$), or large ($ES \geq 0.80$). The level of significance adopted was $P < .05$. The data were analyzed in Statistica (12.0 software; StatSoft, Tulsa, OK).

Results

Values are expressed as mean and SD.

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Table 1 presents the general characteristics of the sample. Randomization showed that 10 subjects started with the CAF session and 12 subjects started with the PLA session.

Table 2 shows the number of repetitions performed, RPE, and PP in the CAF and PLA conditions. For the number of repetitions performed, there was a significant interaction between CAF and PLA ($F = 18.45$; $P = .02$) with the Tukey post hoc test identifying a significant difference between the first and second sets. The RPE analysis demonstrated no significant intergroup interaction. However, there was a significant intragroup interaction in CAF ($F = 14.37$; $P = .04$) and PLA ($F = 16.65$; $P = .03$) conditions. RPE scores were significantly lower after the first set of CAF and PLA in relation to the other sets. For the PP analysis, there was significant interaction between CAF and PLA ($F = 23.78$; $P = .01$). The PP results demonstrated a progressive increase in the CAF condition from the first to third sets. Conversely, in the PLA condition, PP was only significantly different in the first set compared with the second and third sets. Furthermore, PP was significantly different between CAF and PLA in the second and third sets.

The analysis of blood lactate, by Student *t* test, showed no significant differences in resting values between the CAF (2.1 [0.3] mmol·L^{-1}) and PLA (2.2 [0.2] mmol·L^{-1}; Cohen *d* ES = 0.39). After the end of the exercise, the values were significantly higher ($P = .002$) than those observed at rest, and a significant difference ($P = .03$) was observed between the CAF (7.8 [1.1] mmol·L^{-1}) and PLA (6.0 [0.9] mmol·L^{-1}; Cohen *d* ES = 1.79) conditions.

Table 3 shows the 2-way analysis of variance results for the total number of repetitions performed between subjects, who correctly determined which condition was the CAF condition and which the PLA. Nine subjects correctly identified the CAF trial (true positive) and also performed more repetitions in this condition than after PLA intake. Similarly, those failing to correctly identify the CAF trial also performed more repetitions than after taking the PLA. However, there was a significant difference in the number of repetitions completed after CAF intake between those who identified and those who did not correctly identify CAF ($P < .05$). No differences were observed for RPE, PP, or lactate.

Figure 1 presents the individual responses between CAF and PLA in terms of the total number of repetitions performed.

Discussion

The main findings of the present study were (1) CAF intake increased both the number of repetitions performed and capillary blood lactate while also reducing pain sensation in knee extension exercise with BFR, and (2) subjects who accurately interpreted the CAF trial as such performed more repetitions than those who did not perceive it accurately. In both cases, the Cohen *d* ES were large, ratifying the significant level identified. No study to date has

Table 2 Results of the 2-Way ANOVA With Repeated Measures for Effect of Caffeine Supplementation on the Number of Repetitions, Rating of Perceived Exertion, and Pain Perceived

	Caffeine	Placebo	Intergroup *P* (caffeine vs placebo)	Cohen *d* effect size	Cohen *d* classification
Repetitions					
1st set	28.3 (5.3)*	23.7 (3.2)*	.005	1.05	Large
2nd set	11.6 (3.1)**	8.9 (2.9) **	.03	0.90	Large
3rd set	4.6 (3.6)	2.4 (3.0)	NS	0.66	Medium
Total	44.5 (9.4)	35.0 (6.6)	.001	1.17	Large
Rating of perceived exertion					
1st set	6.3 (1.5)*	6.1 (1.9)*	NS	0.11	Small
2nd set	8.4 (1.1)	8.5 (1.5)	NS	0.07	Small
3rd set	9.4 (0.3)	9.6 (0.8)	NS	0.33	Small
Pain perceived					
1st set	5.2 (1.0)*	5.9 (2.2)*	NS	0.40	Small
2nd set	6.9 (1.5) **	8.4 (1.4)	.04	1.03	Large
3rd set	8.7 (0.4)	9.5 (0.6)	.01	1.57	Large

Abbreviations: ANOVA, analysis of variance; NS, nonsignificant difference.
*Intragroup significant difference ($P = .02$) from first and second sets. **Intragroup significant difference ($P = .04$) from third set.

Table 3 Results of the 2-Way ANOVA for the Total Number of Repetitions Performed Between Subjects Who Scored Correctly and Those Who Made a Mistake About Caffeine Intake

	Caffeine	Placebo	Intergroup *P* (caffeine vs placebo)	Cohen *d* effect size	Cohen *d* classification
Correct identification of caffeine (n = 9)	48.8 (4.8)	35.7 (7.7)	.002	2.04	Large
Incorrect identification of caffeine (n = 13)	41.5 (7.1)	34.5 (6.0)	.001	1.06	Large
Intragroup *P* value (correct vs not correct)	.04	NS	—	—	—

Abbreviations: ANOVA, analysis of variance; NS, nonsignificant difference.

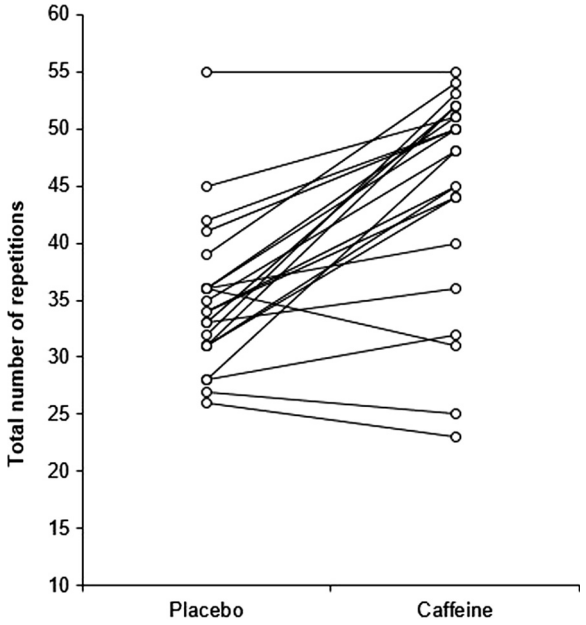

Figure 1 — Individual responses between caffeine and placebo sessions.

examined the concurrent effects of CAF ingestion and BFR on strength performance, and few prior studies examining CAF ingestion have also examined whether the participant's perception of the substance ingested influences the response to the subsequent exercise protocol. As such the results of the present study are novel and extend the literature pertaining to effects of CAF ingestion on exercise performance.

One of the possible mechanisms of action of CAF on physical performance occurs through the increase in the release of calcium in the sarcoplasmic reticulum, boosting the excitation–contraction process.[13] However, it is possible that calcium availability may be impaired under hypoxia conditions.[14] Thus, one of the potential explanations is that exercise with BFR reduced the availability of calcium and, consequently, compromised performance. In the present study, as performance did not decrease, on the contrary, it increased; we suggest 2 hypotheses to explain the seemingly paradoxical findings we present. First, a BFR threshold of 80% BFR was employed in the present study as this threshold is commonly used in the literature.[19] As a consequence, the experimental model did not apply vascular occlusion that interrupted 100% of the blood flow. This means, in the current study, there was no total hypoxia condition, despite less oxygen availability. Thus, even assuming a lower availability of calcium, this may not have been sufficient to compromise performance. Second, the release of calcium into the sarcoplasmic reticulum is not the only mechanism of action by which CAF influences performance.[20] Given that there is debate in regard to the mechanism by which CAF is ergogenic, the results of the current study would imply that increased CAF availability in the sarcoplasmic reticulum might not be the prime mechanism by which CAF ingestion enhances muscular performance.

It has previously been established that CAF can also act in the central nervous system by blocking adenosine receptors,[21] attenuating the action of adenosine and increasing the release of adrenergic neurotransmitters to reduce PP.[22] In the context of the current study, this may have helped individuals to continue performing the exercise for a longer duration and, consequently, increasing the number of repetitions performed with a concomitant increase in blood lactate values postexercise. The observed increase in blood lactate might also be associated with increased PP. However, in the present study, PP was lower in the CAF condition (with large Cohen *d* ES) throughout the sets. In addition, lactate appears to influence PP depending on the concentration of protons and ATP (adenosine triphosphate).[23] Thus, high lactate values may not increase PP. This finding aligns with other studies that have also reported dampened PP during resistance exercise with CAF ingestion.[11] Conversely, no significant differences in RPE were identified as a consequence of the substance ingested. Such a finding is congruent with other resistance exercise data, where the authors hypothesized that CAF may be able to improve performance by maintaining similar levels of perceived exertion to those who produced less work.[24]

Concerning the number of repetitions during exercise with BFR, there was a significant difference between CAF and PLA until the second set ($P < .05$ and large Cohen *d* ES). The nonsignificant difference in the third set may be associated with fatigue in both groups. In this context, the ergogenic effect of CAF may not be realized when several sets are performed to exhaustion and could be indicative that CAF enhances peak strength performance. It is important to note that there was a significant difference in repetitions between subjects who correctly identified CAF intake and those who did not identify the intake of CAF. Some studies have carried out an individual analysis on the effects of CAF on performance and identified that some subjects did not present an ergogenic effect.[25] In other studies, the authors analyzed the side effects of CAF action to see if the sample could identify the substance ingested.[26] In the present study, we did not perform individual performance analysis and also did not verify side effects—we only asked the subjects at the end of the experiment if they thought that CAF had been the first or second capsule consumed. It is important to reinforce that, irrespective of whether subjects correctly identified the CAF ingestion trial or not, the participants performed more repetitions when they ingested CAF compared with when PLA was ingested. However, subjects who correctly identified CAF performed more repetitions than those who did not. Our results agree with the study by Saunders et al.[27] The authors found that cyclists who correctly identified CAF-improved cycling performance to a greater extent than the overall effect of CAF, and the performance also improved when participants ingested CAF while they believed they were ingesting PLA. Therefore, the results of this study reinforces the possibility that CAF has an individualized physiological and psychological action, allowing some subjects to have a superior ergogenic effect to others as a consequence of their expectancy of the effect of the substance they have ingested.

In regard to blood lactate responses to exercise, we observed an increase in this variable after exercise in both the PLA and CAF conditions. Studies have shown that resistance exercise with BFR may result in blood lactate concentration values similar[5] or higher[28] when compared with high-intensity exercise without BRF. It is worth mentioning that this type of exercise has some peculiarities in relation to conventional training of moderate/high intensity, among them is maintaining the blood flow restriction even in the recovery periods, which can significantly decrease the removal of blood lactate in the target muscles.[6]

However, in the current study, blood lactate was higher after the exercise performed with CAF intake. Such a finding is congruent with recent meta-analytical data identifying that acute CAF intake significantly increases plasma lactate.[29] In the present study,

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a possible explanation of the greater number of repetitions performed after CAF ingestion, would have also been associated with a longer duration of effort and a longer time in the BFR condition. As a consequence of this, blood lactate may have been elevated simply because of the greater work performed in the CAF condition rather than because of the CAF ingested. Despite this potential explanation, some studies have shown that plasma lactate did not change after CAF intake even with increased performance.[30] In this sense, the effect of CAF on plasma lactate is still inconclusive and requires further investigation.

Practical Applications

The present study is the first to investigate the acute effects of CAF on an exercise performed with BFR. The positive effect of acute CAF intake to increase the number of repetitions during unilateral knee extension with BFR may help practitioners, athletes, and coaches to optimize the performance in this model of resistance exercise. Notwithstanding, regardless of the results presented, there are some limitations of the current study. We used a single unilateral lower limb exercise as our outcome measure of resistance exercise performance. The results of the present study are only reflective of this type of exercise and we cannot confirm that the results presented herein would be reproducible in bilateral exercises and/or with different muscle mass. Plasma CAF concentration was not measured and thus, we cannot confirm the bioavailability of this substance in all study subjects. We did not test the reliability of the measurements during the exercises protocols. Although our subjects were familiar with resistance exercise protocols, they were not regular practitioners of resistance exercise with BFR. Exercise with BFR may feel different to exercise without BFR and as such some of the observed changes might be attributable to the feeling of the exercise, the error of measurement, or to learning effects. Finally, as this is an acute study, we cannot verify whether the greater number of repetitions performed after CAF intake would be significant to promote a superior effect on strength or muscle mass when analyzed in the long term. Future research would be welcome in order to examine these issues and also seek to replicate the results of the current study.

Conclusion

Acute CAF supplementation increases physical performance and decreases PP in an exercise session of unilateral knee extension with BFR.

Acknowledgments

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